NUS National University of Singapore | ISS

# GC Architecting AI Systems

## Practice Module Briefing

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# Agenda

- Objectives
  - To brief the participants of the graduate certificate on the requirements, conduct and assessment of the practice module
- Topics
  - Requirements of Graduate Certificate
  - Requirements of Group Project
  - Project Schedule
  - Selection of Group Project
  - Project Proposal
  - Typical Examples of Group Project
  - Progress Report for Group Project
  - Sample Group Project Report Template
  - Sample Individual Project Report Template
  - Assessment of Graduate Certificate



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# Requirements of Graduate Certificate - 1

**Objective**: In this graduate certificate, participants will learn to apply essential practices in Explainable and Responsible AI, AI-specific cybersecurity, Agentic AI design patterns, advanced software architecture skills, and MLSecOps/LLMSecOps to design, architect, implement and deploy AI systems that are responsible, ethical, explainable, robust, highly available, reusable, maintainable, and extensible, along with the development of relevant project artifacts.

**Outcomes:** Demonstrate **competency** in all **four** course modules

| **Module 1:** Explainable and Responsible Artificial Intelligence **(3 Days)** | **Module 2:** AI and Cybersecurity **(3 days)** | **Module 3:** Architecting Agentic AI Solutions **(4 days)** | **Module 4:** Integrating and Deploying AI Solutions **(3 days)** |
|---|---|---|---|
| The main objective of this course is to equip participants with practical skills to design and implement AI systems that are accountable, transparent, fair, and ethically responsible. It also aims to develop their ability to assess model fairness, apply bias reduction strategies, use Explainable AI tools, and apply responsible AI frameworks across real-world scenarios. | This course provides cybersecurity and digital professionals with the knowledge and skills to identify vulnerabilities in AI systems, evaluate AI's role in cybersecurity defense, and understand AI-powered attack techniques. Participants will also learn to formulate security strategies that address the risks in AI-based systems. | By combining hands-on experience with system-level thinking, this course prepares professionals to lead the architecture, design and implementation of next-generation AI solutions driven by autonomous, multi-agent systems. | This course provides comprehensive coverage on the principles, methods, and tools for integrating and deploying real-world systems that incorporate intelligent components. Participants will learn MLSecOps and LLMSecOps to streamline the entire ML/LLM lifecycle. |



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# Requirements of Graduate Certificate - 2

**Means** of demonstrating competencies in the four courses:

- Demonstrate understanding and mastery of the skills taught in the above course modules in a **group project (100%) - Practice Module**
- **There is no written exam**



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# Practice Module Prerequisites

- Participants must have successfully completed all (or have been exempted from) the four course modules for the Graduate Certificate in Architecting AI Systems.



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# Project Schedule

**Module Briefing**

**Dec 25-Jan 26**

**15 Jan 26**

**Form a team**

Participants are to form a group of 4-5 members by signing up to a Team in Canvas:

- Canvas course: *NUS-ISS-SWE5008-NCS-2026-01*

**Proposal Submission**

**22 Jan 26**

The participants submit their proposals for review by the lecturers; earlier submissions are welcome:

- Submit a **Project Proposal** for your team to assignment *Project Proposal* in Canvas

**26-30 Jan**

**Proposal Review**

The lecturers provide the review comments. The participants revise the proposals, if necessary, and kick start their projects

**Project Conduct/Progress Report**

**1 Feb 26 - 3 Apr 26**

The participants conduct their projects and consult the lecturers on issues. Each group submits a fortnightly progress report for review by the lecturers:

- Submit a **Progress Report** for your team to assignment *Progress Report* in Canvas
- Report name convention: Team X_<date>

**6 Apr 26 - 10 Apr 26**

**Project Presentation**

The participants present their projects

**Report Submission**

**17 Apr 26**

The participants submit their project report, incorporating remarks gathered during their project presentations.

- Submit a **Group Project Report** for your team to assignment *Project Report (Group-Based)* in Canvas
- Submit an **Individual Project Report** to assignment *Project Report (Individual)* in Canvas

**Early May 26**

**Peer Assessment**

***Some dates are tentative**



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# Selection of Group Project

Team size: 4-5 members

An estimated **15 days** of effort by each participant

- Not contiguous and stretch over **several weeks**

Participants are to **source** for suitable projects, either from within your **own organizations**, your **own ideas** or from **other approved sources**

A **suitable** project is one that adequately **utilizes and demonstrates** the skills learned in the course modules

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# Project Proposal

- The proposal should include the following content

| Section | Content |
|---|---|
| Project Title | Title |
| Project Sponsor | Name, title, contact (if applicable) |
| Project Members | Names |
| Overview | Describe the context and the business problem solved by the Agentic AI system |
| General Architecture | Outline the general architecture of the AI system. This can be a logical architecture if deployment details are not finalized. The architecture should set the context for your defined scope. |
| Scope of Work | Describe the agents in your system. Explain how your implementation will demonstrate key considerations, eg assurance, trust, fairness, accountability, ethical issues, explainability, governance, security measures, agent autonomous and orchestration, as well as MLOps and LLMOps practices. |
| Effort Estimates | List the rough WBS tasks and their estimated efforts. This is to ensure that you have thought about the approach and the implementation effort |



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# Questions to Consider

- What are the **key pain points** in current workflows that multi-agent system can address? What **additional value-creation opportunities** arise specifically from introducing a multi-agent AI system, even beyond addressing today's workflow gaps? Eg. in telemedicine platform, could autonomous agents proactively identify hidden risks (e.g., medication interactions)?
- Justify the **pros and cons** for such agentic AI system adoption or implementation?
- Who are the **key stakeholders** of this project?
- How should the **agents coordinate** to ensure a seamless user experience
- What are the **expected benefits** of using a modular, agentic AI architecture versus a monolithic chatbot or simple rules-based automation? (e.g., reduced workload, better patient guidance, flexibility, explainability, traceability)
- How can these **benefits** be **demonstrated** during the project demo or presentation? (e.g., scenarios where agents show autonomy, handle queries responsibly, escalate appropriately)
- What **scale** (e.g., concurrent patient sessions) is the solution designed to support?
- How can this scalability claim be demonstrated or **justified**? (e.g., load tests, architectural scalability characteristics)
- How does the solution ensure **explainability** and **traceability** of agent decisions?
- What **safeguards** are in place to mitigate bias, ensure fairness, accountability, trust and assurance during deployment?
- What common services (e.g., shared memory, logs) will be required to support these agents?
- What reusable libraries or **frameworks** (e.g., LangChain, LangGraph) are leveraged for agent orchestration?
- What are the **AI-specific security risks** (e.g., prompt injection, hallucinations, adversarial prompts) and how are they mitigated?
- How does the design ensure **safe** behaviour even under malicious or unexpected inputs?
- Which part of AI system lifecycle can be **automated** and how it can reduce the development effort and improve system quality attributes?
- Is the amount of **effort** needed to do the implementation still within the guideline?


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# Typical Example of Group Project - 1

Project: Care-AI telemedicine enhancement multi-agent system

Problem description:

- CARE-AI is a telemedicine enhancement initiative for a regional virtual clinic that aims to streamline pre-consult triage, improve patient education, and reduce repetitive workload for healthcare staff. The proposed solution involves a multi-agent system, where specialized AI agents collaborate to support patients before and after online consultations.
- Key agent roles and system features include:
  - Triage Agent – Collects symptoms and classifies urgency level
  - Preparation Agent – Guides patients in preparing for their consultation (e.g., blood pressure logs, medication list)
  - Education Agent – Answers common questions using vector-based medical information retrieval
  - Follow-up Agent – Provides post-consultation advice and reminders (e.g., medication, lifestyle guidance)
  - Escalation Agent – Monitors conversations and flags safety-critical cases for human intervention
- Agents may coordinate using shared memory or messaging. Each agent is expected to demonstrate autonomy through reasoning, planning, and tool use. The system should also maintain logs to ensure traceability and explainability of agent decisions.
- The project may use simulated data and is not expected to be production-ready. The emphasis is on demonstrating agentic behaviour, explainable and responsible AI practices, AI-specific security considerations, and sound engineering principles including thoughtful architecture, modular design, and MLSecOps/LLMSecOps practices for integration, testing, and monitoring.



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# Typical Example of Group Project - 2

Success criteria:

- **Presentation slides** clearly outlining the problem context, agent roles, system overview, and demo walkthrough
- **System architecture document** providing a high-level view of the overall system. This should cover the logical and physical architecture, infrastructure setup, service deployment strategy, integration points, data flow across components, and justification of architectural styles and technologies used.
- **Agent design documentation** describing the internal logic and behaviour of each agent. This includes the reasoning patterns used, planning loops, memory mechanisms, tool integrations, and how autonomy is achieved. Also specify prompt patterns, fallback strategies, and inter-agent communication protocols.
- **Explainable and Responsible AI report** explaining for example, how fairness, bias mitigation, explainability, and governance principles were considered and applied. Show how the different stages are aligned with the explainable and responsible AI principles.
- **AI security risk register** identifying risks resulting from potential vulnerabilities (e.g., prompt injection, hallucination), including AI-Agent related risks, and outlining **mitigation strategies** and appropriate security controls.
- **MLSecOps / LLMSecOps pipeline design** specifying tools and workflows for CI/CD, automated testing (including AI security tests), model versioning, monitoring, and logging.
- Well-structured **source code** repository showcasing modular agent implementation with clear separation of logic and well-documented code.
- **Testing artifacts** including unit tests for agent behaviour, end-to-end flow verification and relevant AI security tests.
- Simple **UI prototype** (e.g., web-based or form-based) to demonstrate user interaction and multi-agent orchestration.



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# Progress Report for Group Project

- Submit team progress report to assignment *Progress Report* in Canvas **fortnightly** for review by NUS-ISS
- Report name convention: Team X_<Date>
- The progress report should at least include the following content

| **Section** | **Content** |
|---|---|
| Project Title | Title |
| Date | Date of report |
| Summary of Work Done | List of tasks conducted since last report |
| Effort Expended | Hours expended by each member since last report |
| Problems Encountered | Issues that might need assistance from NUS-ISS lecturers |
| Plan for the Next 2 Week | List of tasks planned for the next 2 weeks |


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# Sample Group Project Report Template

Project title, Team number and members

**1. Executive Summary**

- Brief overview of project objective and scope
- Key highlights of the solution
- Constraints/assumptions

**2. System Overview**

Description of overall solution and how agents work together
High-level workflow diagram

**3. System Architecture**

- Logical architecture diagram with description
- Physical architecture diagram with infrastructure details
- Deployment strategy (e.g., containerization, cloud services)
- Data flow and integration points
- Justification of architectural styles and tech choices

**4. Agent Roles and Design**

For each agent:

- Purpose and responsibilities
- Reasoning patterns
- Planning and memory mechanisms
- Tools used
- Communication protocols and coordination mechanisms

**5. Explainable and Responsible AI Practices**

- How the different stages of development & deployment are aligned with the explainable and responsible AI principles.
- Approach to fairness, bias mitigation, explainability
- Governance framework alignment (e.g., IMDA's Model AI Governance Framework)

**6. AI Security Risk Register**

- Table of identified risks (e.g., prompt injection)
- Mitigation strategies and security controls

**7. MLSecOps / LLMSecOps Pipeline**

CI/CD diagram with description of:

- Automated testing (including AI security tests)
- Versioning and tracking
- Deployment strategy
- Monitoring and alerting
- Logging and auditability

**8. Testing Summary**

- Types of tests performed (unit, integration, security)
- Results

**9. Reflection**



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# Sample Individual Project Report Template

Project title, Team number, Individual name, Agent responsible for (e.g., “Follow-up Agent”)

**1. Introduction**

- Brief overview of the agent’s purpose

**2. Agent Design**

- Description of agent role and functionality
- Reasoning and planning workflow
- Memory and state management
- Tools integrated
- Communication and coordination logic
- Prompt engineering patterns used
- Fallback strategies

**3. Implementation Details**

- Summary of implementation approach
- Code structure overview
- Tech stack of the agent (e.g. LLM model, agent framework, MCP etc.)

**4. Testing and Validation**

- Types of tests implemented eg:
  - Unit tests
  - Integration tests (with other agents or backend services)
  - AI security tests (e.g., adversarial inputs, prompt injection handling)
- Results and key findings

**6. Explainable and Responsible AI Considerations**

- How the different stages of development & deployment are aligned with the explainable and responsible AI principles.
- How this agent addresses explainability
- Bias mitigation strategies (if relevant)
- Handling sensitive content and governance alignment

**7. Security Practices**

- Agent-specific security risks
- Mitigations implemented at code level or workflow level

**8. Reflection**

Personal learning from implementing this agent

Suggestions for future improvement of the agent



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# Peer Assessment

- Peer assessment is conducted in order to **reward** the members who **contribute** more towards meeting the **project objectives**
  - It is **not** a measure of the numbers of **man-hours** expended by the members
- The **Peer Grading System** by NUS-ISS automates the conduct
  - Students are notified and reminded via their **NUS email addresses**
    - Not reading the emails in NUS mailboxes is not a valid excuse
  - Students are **mandated** to rank the contribution of their members with **unique ranks**
    - Additional remarks are welcome
    - Students who **do not rank** by the deadline would be assigned **zero** peer assessment marks



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# Assessment of Graduate Certificate

| Assessment Component | Weight (Company Sponsored) | Weight (Not Company Sponsored) |
|---|---|---|
| **Team based:** | 50% | 50% |
| Project presentation | 20% | 20% |
| Project report | 30% | 30% |
| **Individual:** | 50% | 50% |
| Project report (design of one agent) | 10% | 13% |
| Project report (implementation and testing of the agent) | 10% | 13% |
| Individual reflection | 10% | 14% |
| Peer assessment | 10% | 10% |
| Company sponsor assessment* | 10% | - |
| **TOTAL** | 100% | 100% |

- Presentation, project report and individual report will be graded by a team of NUS-ISS lecturers.
- The participant must pass with a minimum score of 20% (i.e., Grade D) of his/her individual assessment.
- The participant must attain a minimum overall score of 50% (i.e., Grade C) to be awarded the Graduate Certificate in Architecting AI Systems.
- The participant must attain a minimum grade point average of 60% (i.e., Grade B-) across the required graduate certificates to be eligible to be enrolled into a Master of Technology degree.
- Entry into the Master of Technology degree programme is not assured whatever the score attained in a Graduate Certificate, but is solely at the discretion of NUS-ISS.


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# Presentation Guideline

1. **Introduction**
   - Brief overview of project objective and scope
   - Description of overall solution and how agents work together High-level workflow diagram
2. **Overall effort to date**
   - Overall estimate effort versus actual effort of every member till-date

**3. System Architecture**

- Logical architecture diagram with description
- Physical architecture diagram with infrastructure details
- Deployment strategy (e.g., containerization, cloud services)
- Data flow and integration points and design pattern
- Justification of architectural styles and tech choices
- Tech Stack

**4. Agent Design**

- For each agent:
- Purpose and responsibilities
- Planning/reasoning
- Memory mechanisms
- Tools used

**5. Explainable and Responsible AI Practices**

- How the different stages of development & deployment are aligned with the explainable and responsible AI principles.
- Approach to fairness, bias mitigation, explainability
- Governance framework alignment (e.g., IMDA's Model AI Governance Framework)

**6. AI Security Risk Register**

- Table of identified risks (e.g., prompt injection)
- Mitigation strategies and security controls

**7. Application demo**

**8. MLSecOps / LLMSecOps Pipeline and demo**

- Pipeline diagram
- Demo

**9. Testing Summary**

- Types of tests performed (unit, integration, security)
- Results



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# Thank you

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